

Fragile States Index Analysis

Final Project Documentation

Technical and analytical documentation for the Power BI Fragile States Index project.

1. Project Overview

Objective

Build an end-to-end Power BI solution using Fragile States Index data from 2015--2023 to examine current fragility, historical trends, persistent high-risk conditions, underlying indicators, and prioritization signals.

Technology

- Power BI Desktop
- Power Query
- DAX
- Microsoft Excel
- Git / GitHub

2. Source Data

Nine annual FSI files were used:

2015--2023

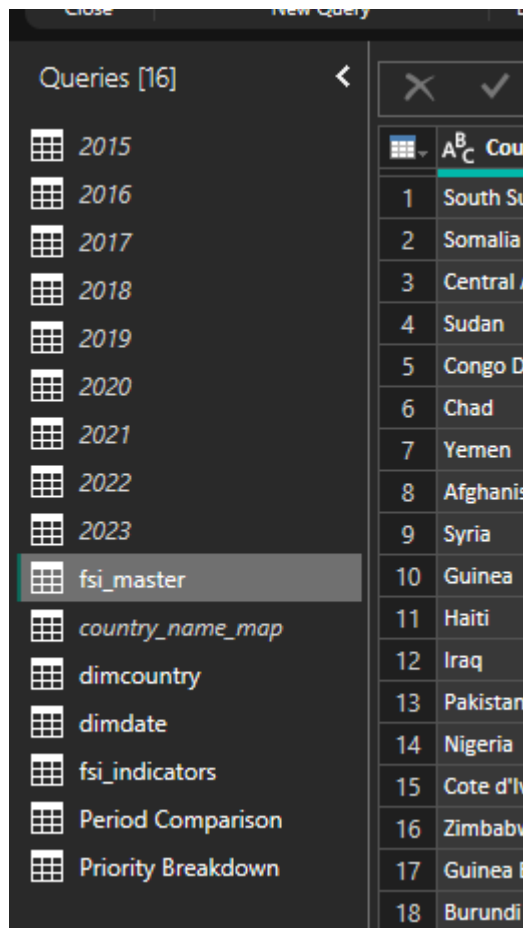
The final analytical grain is:

One row = one country in one year

Each source contains Country, Year, Rank, Total FSI score, and the twelve FSI indicators.

3. Data Preparation

3.1 Source queries



Annual source queries were cleaned and standardized before being appended into `fsi_master`.

3.2 Standardization

Major preparation activities:

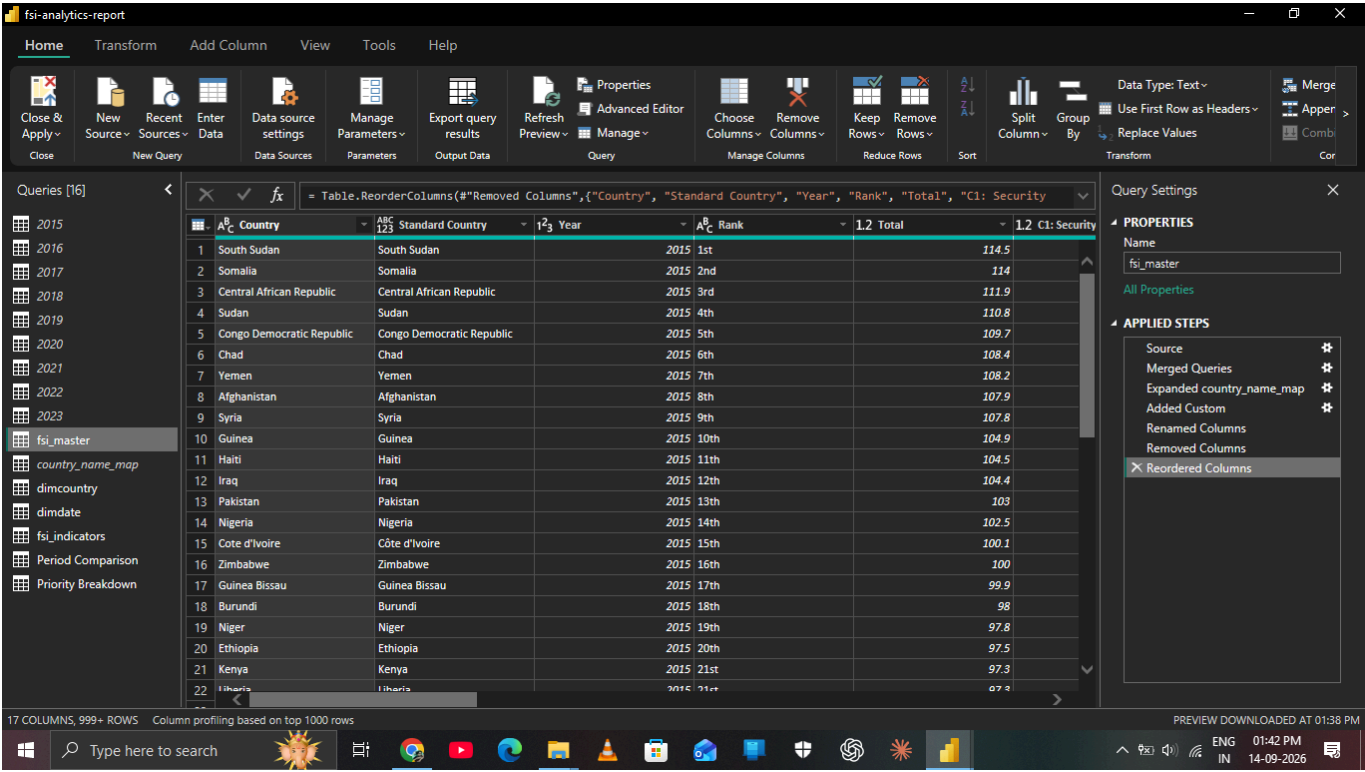
- standardize data types;
- remove unnecessary source fields;
- standardize Year;
- ensure numeric indicator values;
- standardize country names;
- create `StandardCountry`;
- append annual datasets;
- prepare the long-format indicator table.

Known country-name variants were mapped, including Czechia → Czech Republic, Swaziland → Eswatini, Macedonia → North Macedonia, and similar source variants.

The Israel / West Bank change across years was treated as an entity-resolution consideration rather than blindly merging distinct historical structures.

4. Master Dataset

`fsi_master`



The central table contains Country, Year, Rank, Total, the twelve indicators, and StandardCountry.

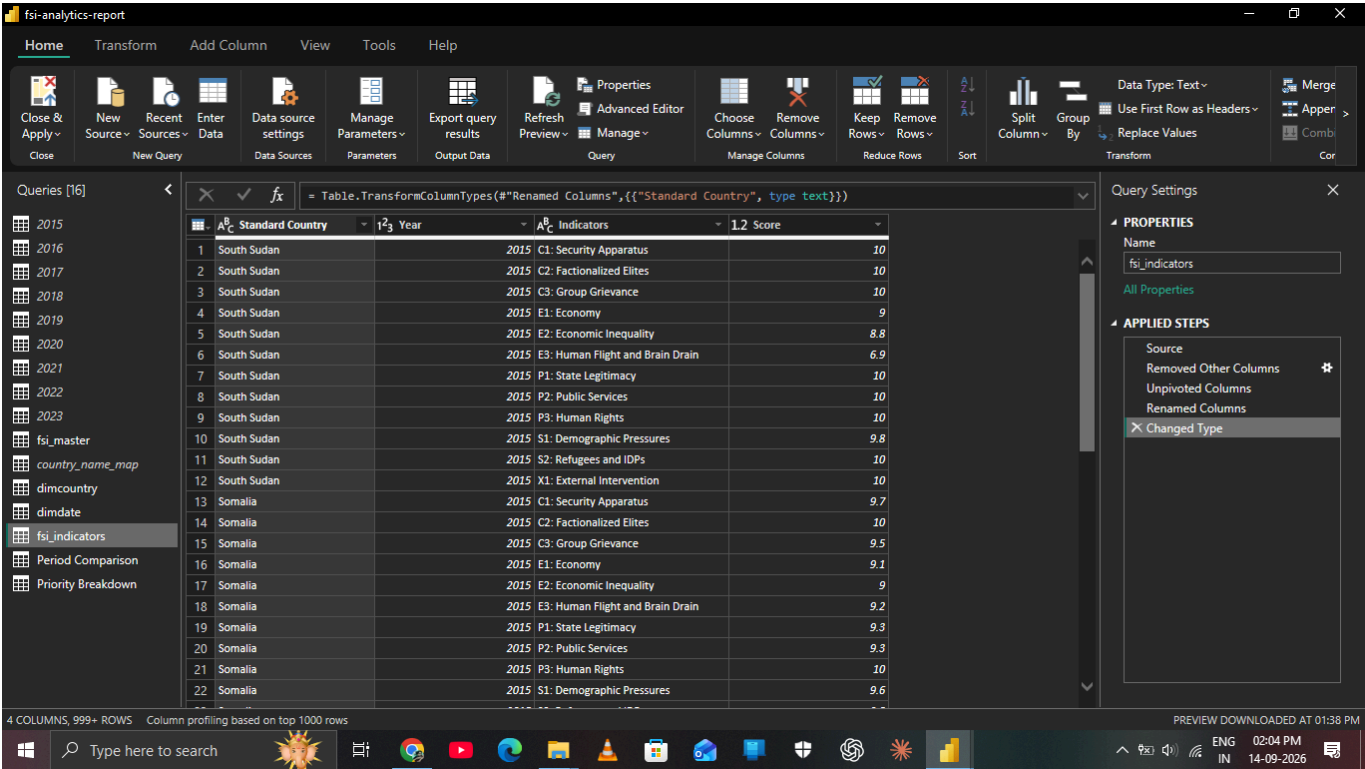
The analytical uniqueness rule is:

StandardCountry + Year = one country-year observation

The final model contains **180 standardized countries** across **9 years**.

5. Indicator Transformation

fsi_indicators



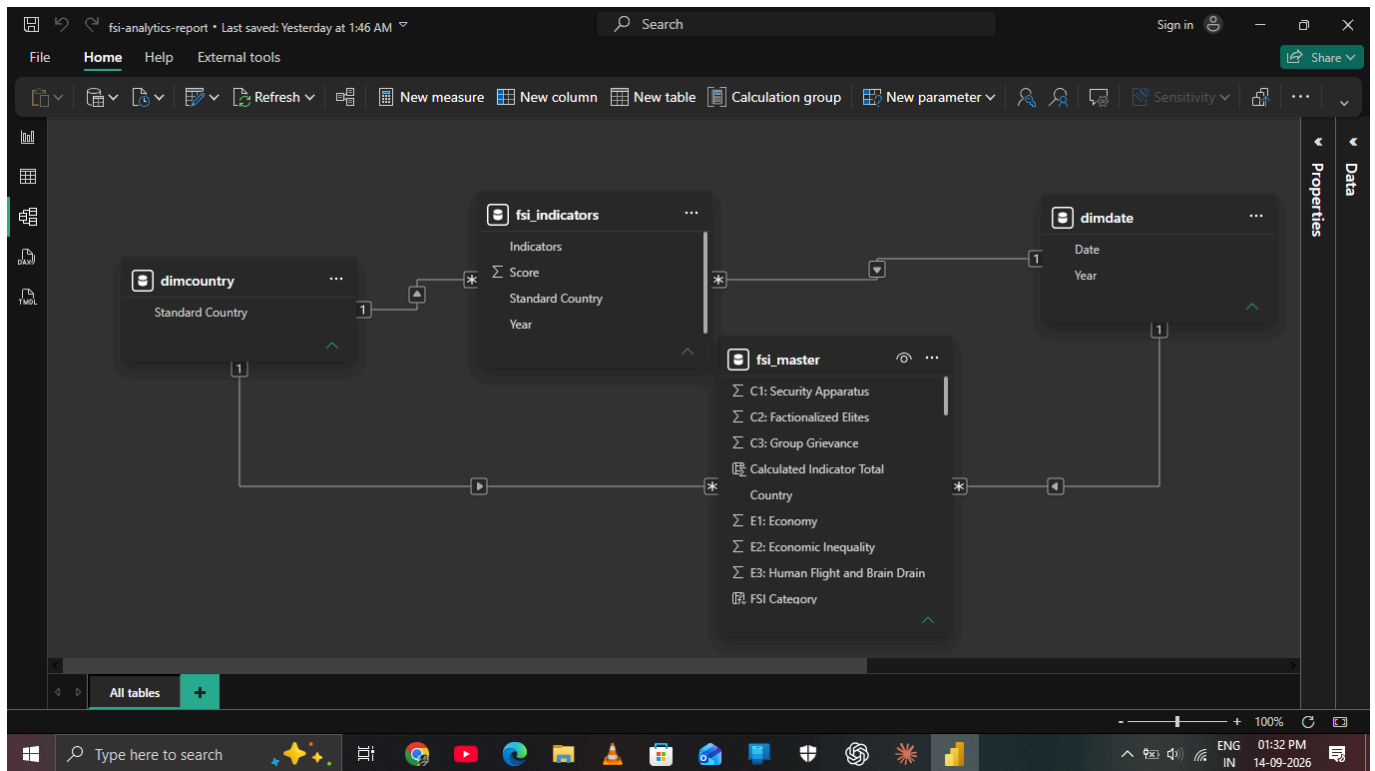
The twelve indicator columns are unpivoted into:

StandardCountry Year Indicator Score

Afghanistan 2023 C1 Security Apparatus score Afghanistan 2023 C2 Factionalized Elites score Afghanistan 2023 C3 Group Grievance score

This structure supports indicator comparisons, trends, profiles, elevated-indicator analysis, and intervention-focused views.

6. Semantic Model



Tables

fsi_master

Primary country-year FSI table.

fsi_indicators

Long-format indicator table.

DimCountry

Unique standardized country dimension.

DimDate

Annual date dimension.

country_name_map

Country-name standardization helper.

Period Comparison

Disconnected table containing Earlier Period and Recent Period.

Priority Breakdown

Disconnected table containing priority signal counts from 3 to 0.

Relationships

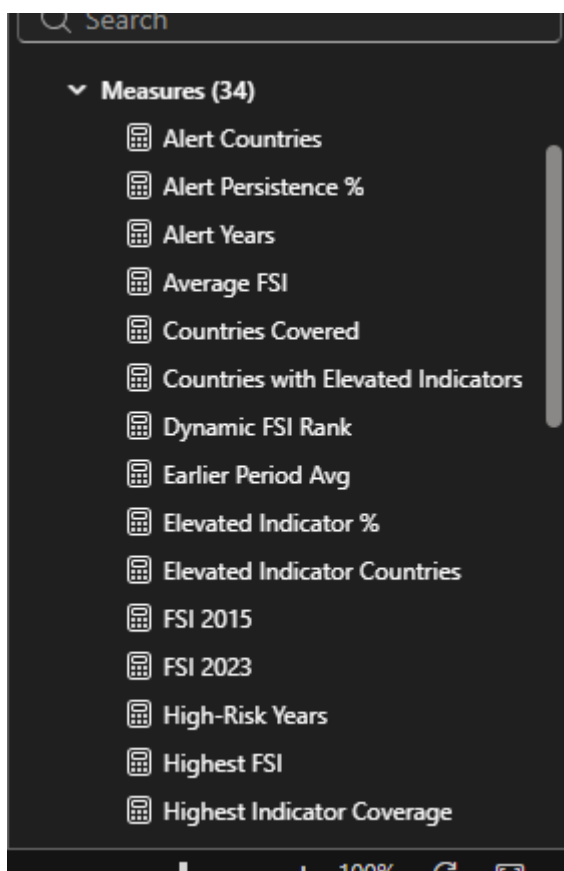
```
DimCountry[StandardCountry] 1 → * fsi_master[StandardCountry]
DimDate[Year]                1 → * fsi_master[Year]

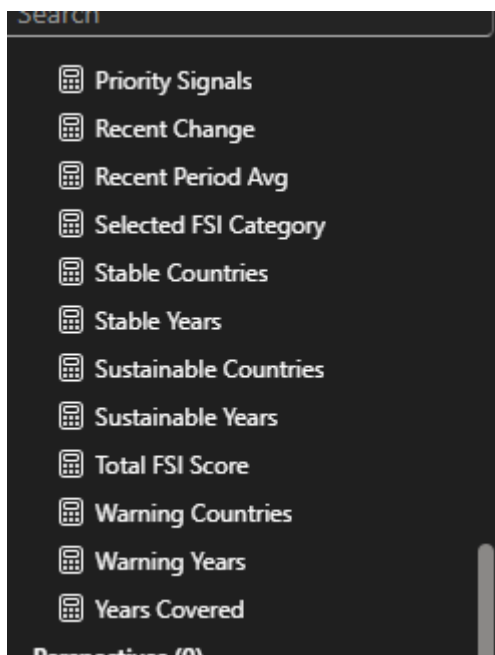
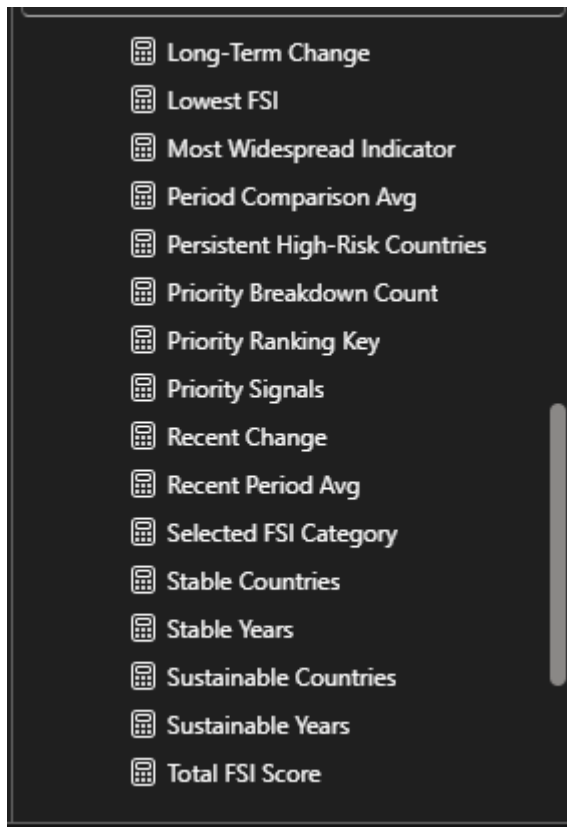
DimCountry[StandardCountry] 1 → * fsi_indicators[StandardCountry]
DimDate[Year]                1 → * fsi_indicators[Year]
```

Relationships are active and single-direction. Fact-to-fact relationships are intentionally avoided.

7. DAX / Analytical Layer

The project uses approximately thirty measures. They are grouped into core metrics, time comparisons, risk/persistence, prioritization, and indicator analysis.





Core measures

```
Total FSI Score = SUM('fsi_master'[Total])
```

```
Average FSI = AVERAGE('fsi_master'[Total])
```

```
Highest FSI = MAX('fsi_master'[Total])
```

```
Lowest FSI = MIN('fsi_master'[Total])
```

FSI category

```
FSI Category =
SWITCH(
  TRUE(),
  'fsi_master'[Total] >= 90, "Alert",
  'fsi_master'[Total] >= 60, "Warning",
  'fsi_master'[Total] >= 30, "Stable",
  "Sustainable"
)
```

A separate sort helper displays categories as:

Alert → Warning → Stable → Sustainable

Time comparison

Long-Term Change = FSI 2023 – FSI 2015

Earlier Period = **2015--2018**

Recent Period = **2020--2023**

The period comparison uses a disconnected **Period Comparison** table and a measure that returns the corresponding period average.

Risk and persistence

High-Risk Years counts Alert/Warning years for the current country context.

Persistent High-Risk Countries counts countries with at least four Alert/Warning years across 2015-2023.

Validated result:

122

Dynamic country ranking

```
RANKX(
  ALL('dimcountry'[StandardCountry]),
  [FSI 2023],
  ,
  DESC,
  DENSE
)
```

Priority signals

Three binary signals are added:

```
Current Risk
+ Persistent Risk
+ Deterioration
= Priority Signals (0-3)
```

Conditions:

- $FSI\ 2023 \geq 60$
- Alert/Warning in ≥ 4 years
- $FSI\ 2023 - FSI\ 2015 > 0$

The **Priority Ranking Key** is a technical Top N ranking mechanism; it is not itself a humanitarian utility score.

Indicator analysis

An indicator score of **6/10 or higher** is treated as elevated.

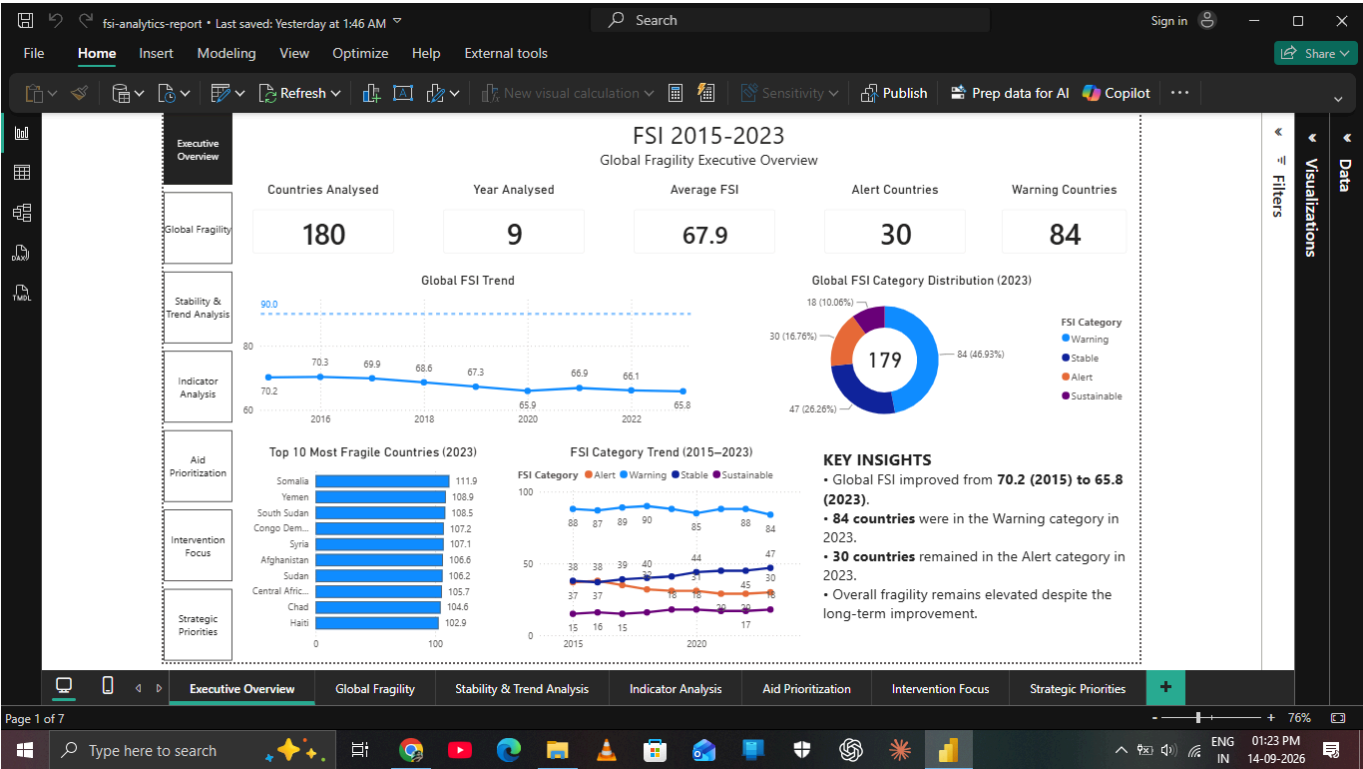
Measures calculate:

- Elevated Indicator %
- Elevated Indicator Countries
- Countries with Elevated Indicators
- Most Widespread Indicator
- Highest Indicator Coverage

The 6/10 threshold is an analytical project choice, not an official FSI classification.

8. Dashboard Pages

Page 1 --- Executive Overview

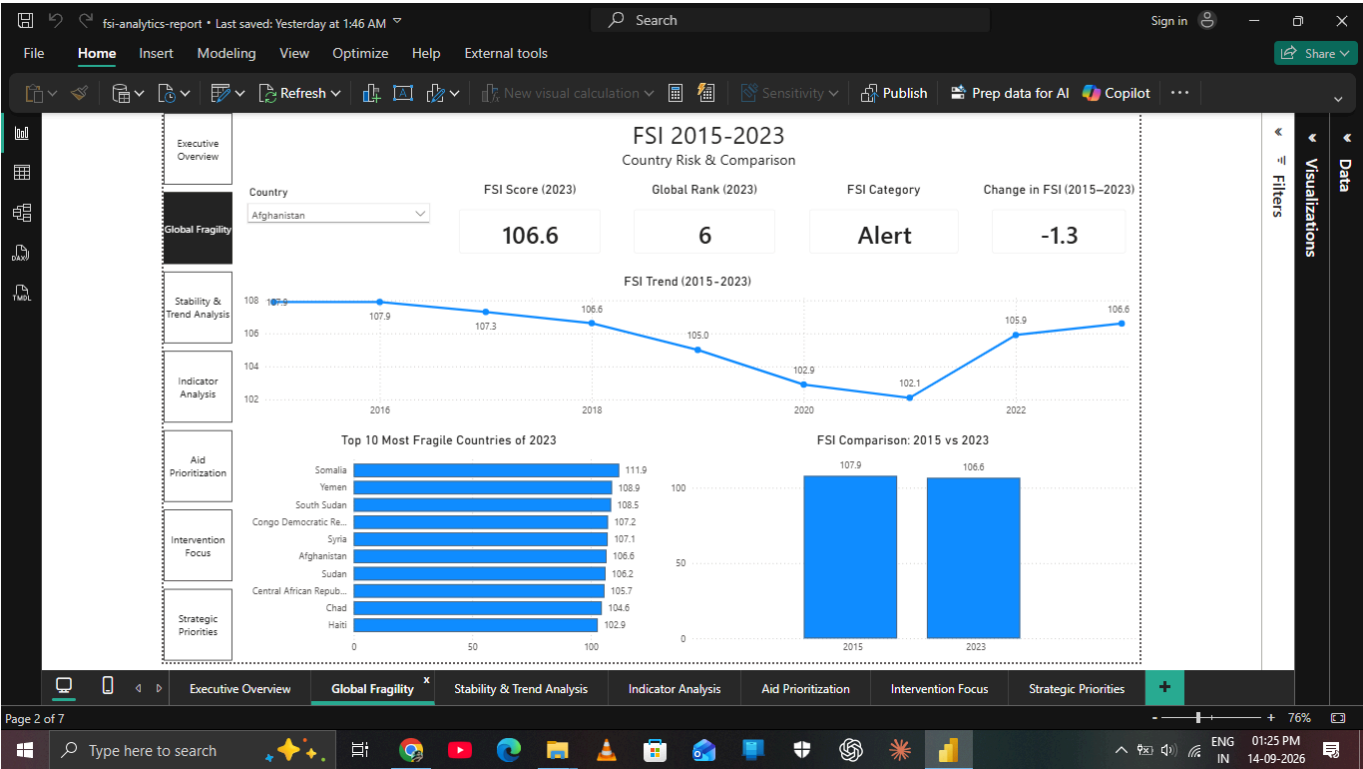


Objective: Establish the global baseline.

Main elements: KPI summary, global FSI trend, 2023 category distribution, Top 10 fragile countries, category trend, key insights.

Interaction: Primarily global/static. Historical context is preserved.

Page 2 --- Country Risk & Comparison

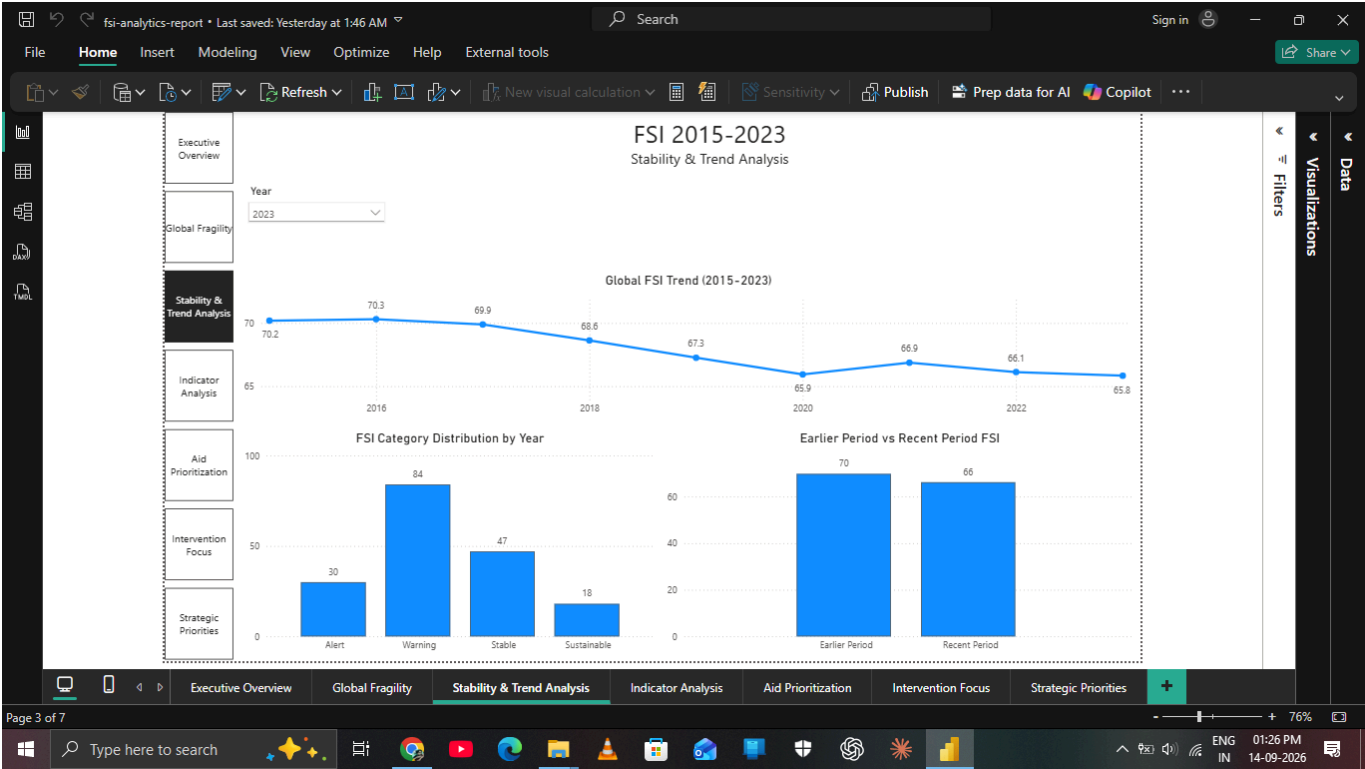


Objective: Assess an individual country against the global benchmark.

Main elements: 2023 FSI, dynamic rank, category, long-term change, country trend, Top 10 benchmark, 2015 vs 2023 comparison.

Interaction: Country slicer controls country-specific visuals while the global Top 10 benchmark remains independent.

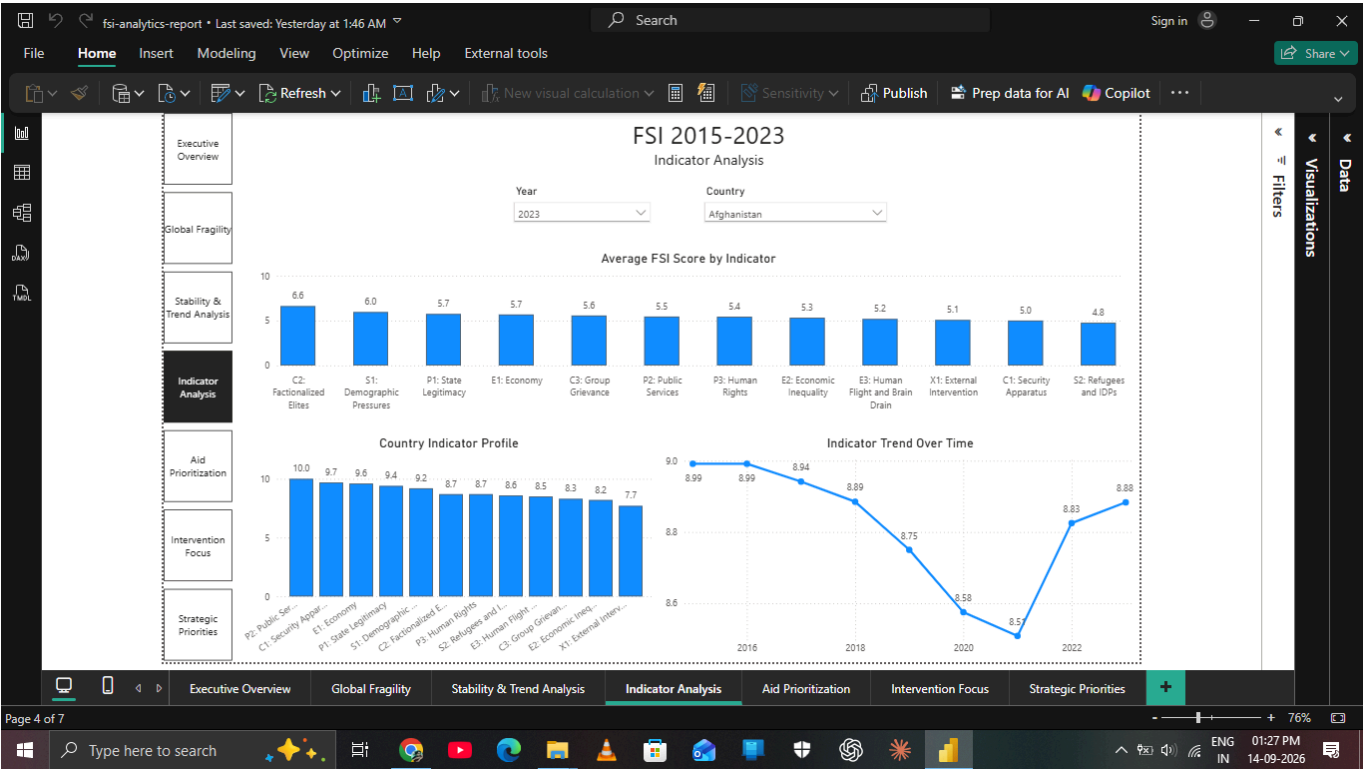
Page 3 --- Stability & Trend Analysis



Objective: Analyze global change and category distribution.

Interaction: Year controls the category distribution. The global trend and fixed-period comparison remain independent so historical context is preserved.

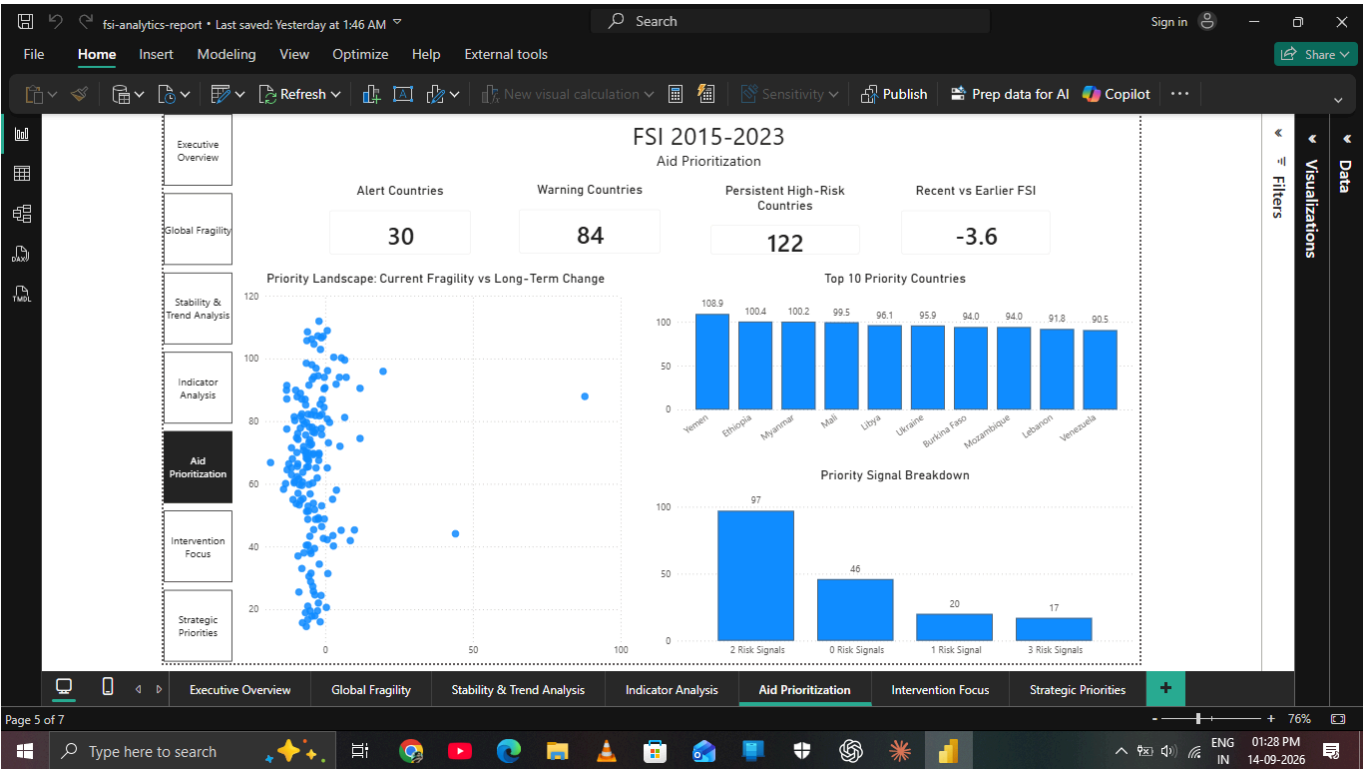
Page 4 --- Indicator Analysis



Objective: Analyze the twelve underlying dimensions.

Interaction: Country and Year slicers control relevant current views. Selecting an indicator updates the historical trend and highlights the corresponding country-profile indicator. The main indicator comparison remains independent of the Country slicer.

Page 5 --- Aid Prioritization

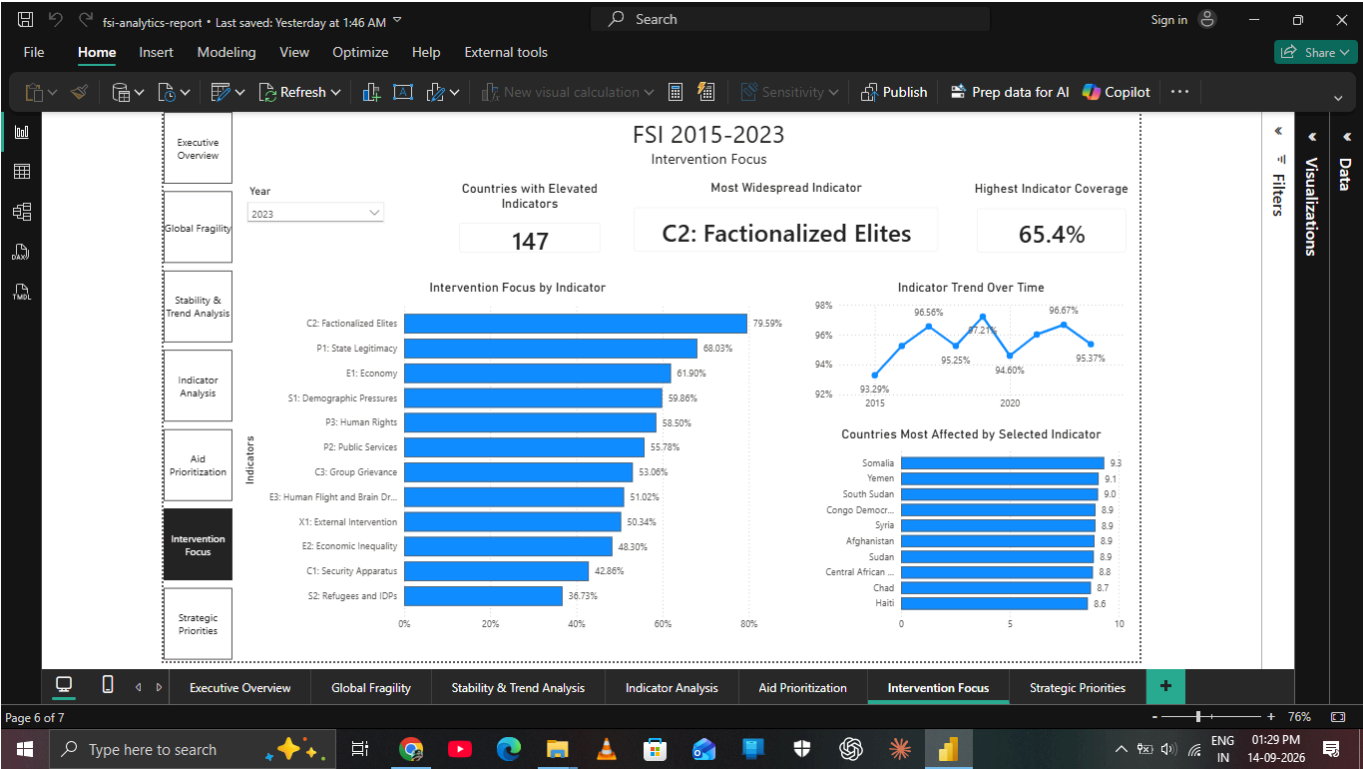


Objective: Identify countries with multiple risk signals.

Main elements: current Alert/Warning counts, persistent high-risk count, period comparison, priority landscape, Top 10 priority countries, signal breakdown.

Interaction: Intentionally global/static; no unnecessary cross-filtering.

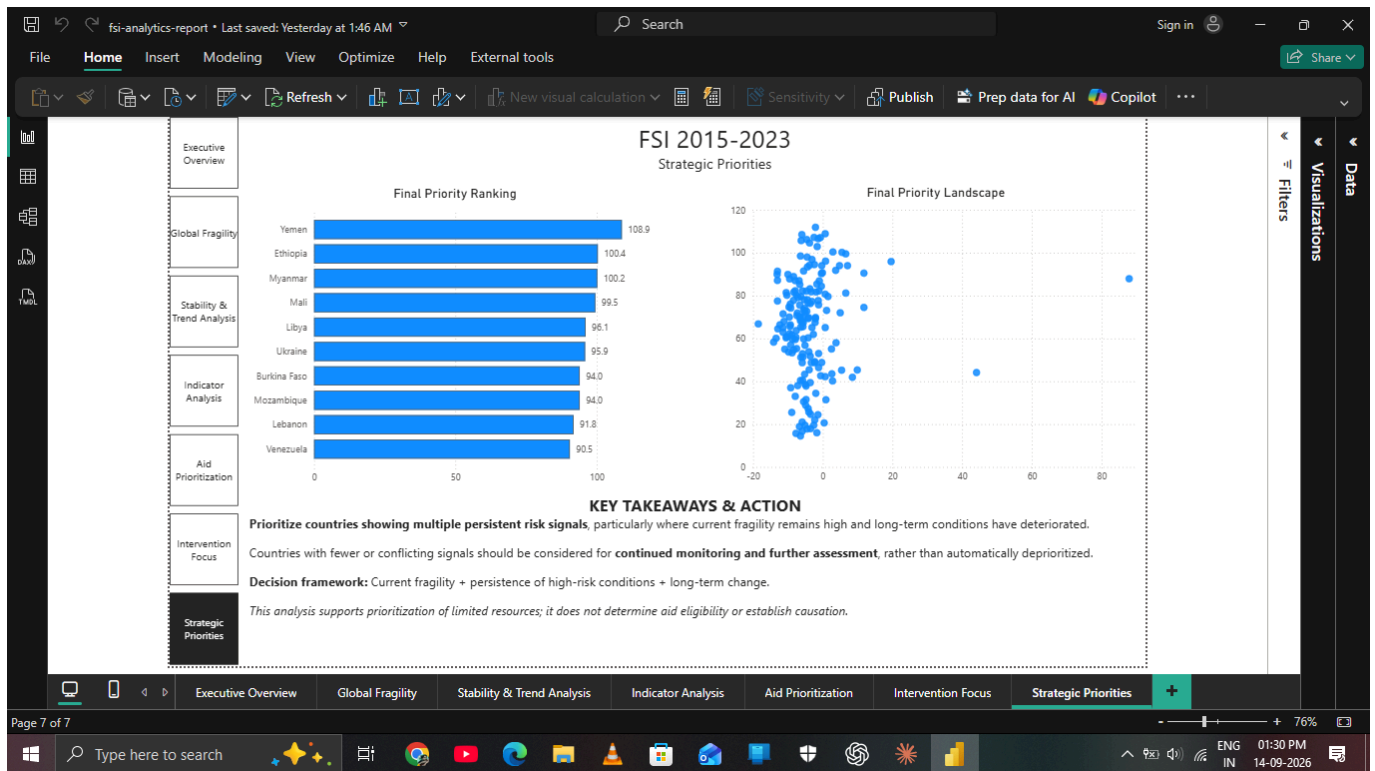
Page 6 --- Intervention Focus



Objective: Identify widespread elevated dimensions and affected countries.

Interaction: Indicator selection updates the trend and affected-country ranking. Year controls current-year views while the trend preserves historical context. KPI cards respond to Year but not indicator selection.

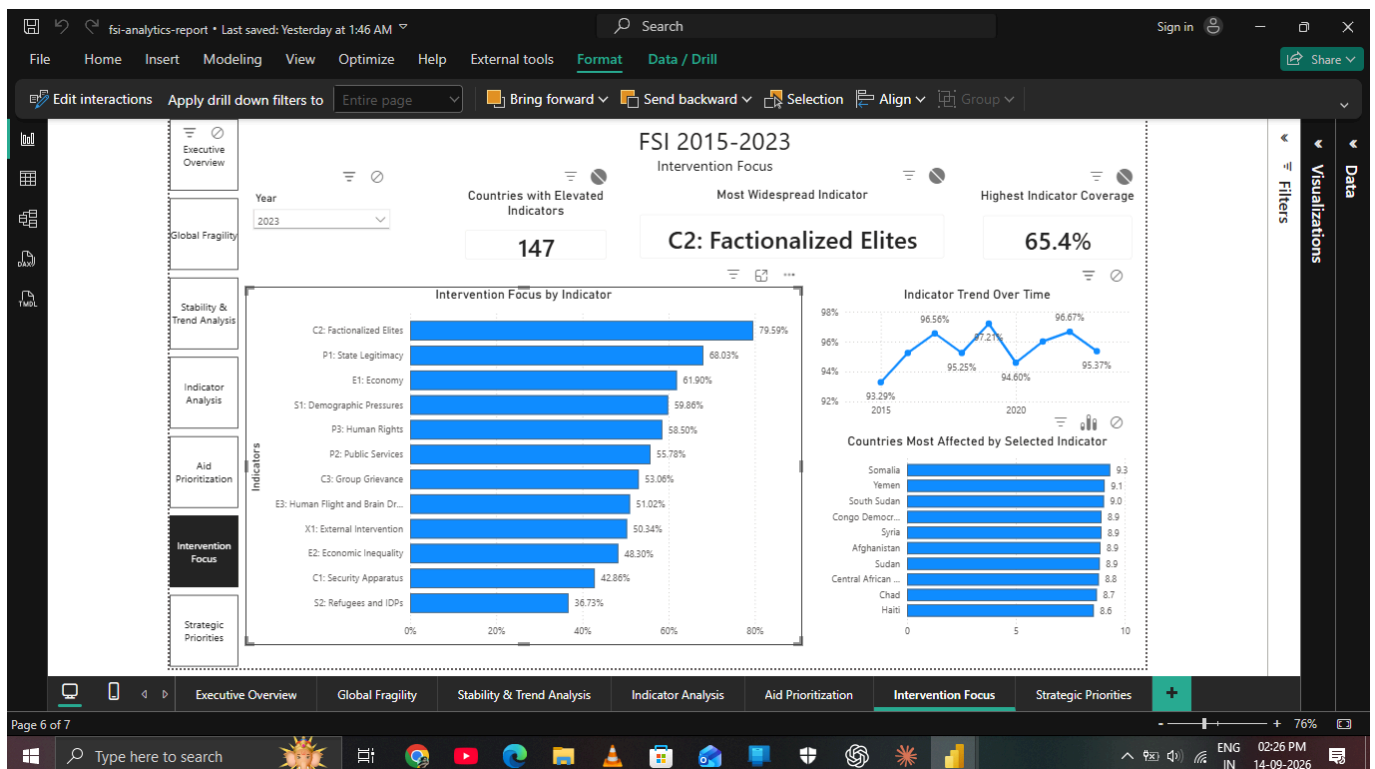
Page 7 --- Strategic Priorities



Objective: Provide the final strategic synthesis.

Interaction: Intentionally static. Ranking and landscape are complementary rather than cross-filtering.

9. Visual Interaction Architecture

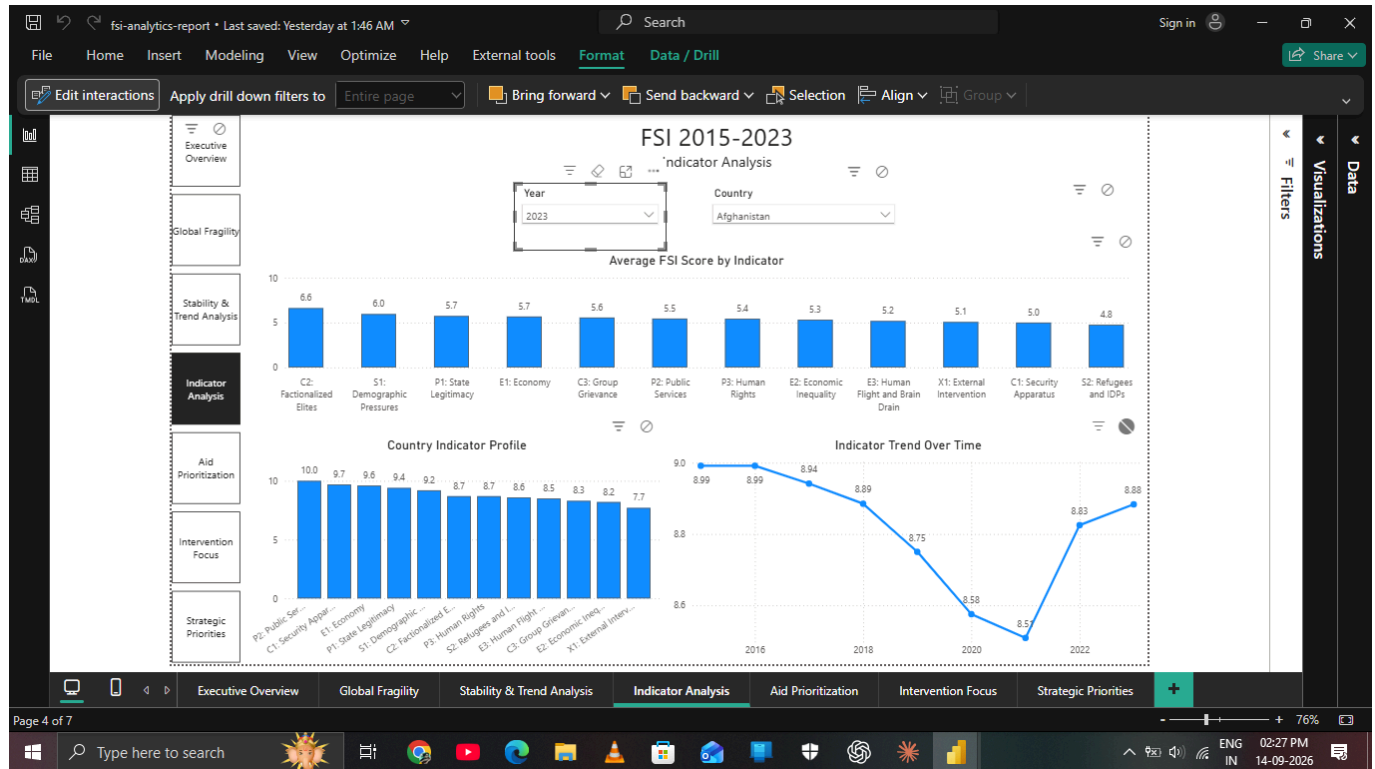


Interactions were treated as part of analytical design rather than simply accepting Power BI defaults.

Principles

- **Preserve context:** historical trends remain available where useful.
- **Filter where detail is required:** country and indicator selections affect relevant visuals.
- **Disable misleading interactions:** global benchmarks remain independent when appropriate.
- **Avoid unnecessary complexity:** visuals interact only where the relationship adds analytical value.

Example



Typical flows:

Select Year → update current-year visuals

Select Indicator → update indicator trend + affected-country analysis

10. QA & Validation

Data validation

- Exact duplicate full rows were checked.
- Indicator values were checked against the expected 0--10 range.
- Year was standardized to Whole Number.
- Country names were standardized.
- Country + Year uniqueness was validated.
- Annual schemas were aligned before append.
- Source ranking anomalies were preserved rather than silently corrected.

Analytical validation

Key outputs were independently checked:

- **180 standardized countries**
- **9 years**
- **2023 category distribution: 30 / 84 / 47 / 18**
- **122 persistent high-risk countries**
- Priority signal breakdown
- Page-level ranking logic
- Page 6 interaction behavior
- Page 7 static interaction configuration

Final report QA

The final audit checked saved slicer states, visual titles, layout, interactions, ranking filters, category sorting, KPI outputs, and historical trend behavior.

Final saved states were corrected so the report opens in the intended latest-year context where appropriate.

11. Key Design Decisions

No opaque composite priority score

Three transparent signals were preferred over a complex weighted score.

No causality claims

The analysis is descriptive/comparative and does not claim that one FSI factor causes another outcome.

No automatic aid eligibility

Priority signals identify analytical priority, not funding eligibility.

Controlled interactions

Interactions were deliberately configured to preserve analytical context.

No unnecessary dynamic narrative

A static final takeaway was preferred over over-engineered dynamic text.

12. Final Project Outcome

```
Source Data
↓
Power Query Preparation
↓
Standardized Master Dataset
↓
Dimensional Semantic Model
↓
```

```
graph TD; A[DAX Analytical Layer] --> B[Interactive Dashboard]; B --> C[Validation & QA]; C --> D[Humanitarian Decision Support];
```

DAX Analytical Layer
↓
Interactive Dashboard
↓
Validation & QA
↓
Humanitarian Decision Support

The completed solution demonstrates an end-to-end workflow covering data preparation, modeling, analytical calculation, visualization, interaction design, validation, and decision-oriented storytelling.

13. Responsible Use

This project is an analytical portfolio exercise using Fragile States Index data. Its prioritization framework demonstrates transparent decision-support methodology. It should not be used as a standalone mechanism for determining humanitarian eligibility, funding allocation, or operational intervention.

Real-world decisions should combine FSI evidence with current humanitarian, socioeconomic, conflict, displacement, health, food-security, demographic, and local contextual information.